A SAMPLE OF SECTION A TYPE QUESTIONS IN ME 366

UNIT 1

Q1. The basic law of heat conduction is known as?

(a) Newton-Rickman's Law (c) Planck's Law

(b) Fourier's Law (d) Stefan-Boltzmann's Law

Q2. Which of the following gases will have the least value of thermal conductivity at the same temperature?

(a) Argon (c) Nitrogen

(b) Helium (d) Carbon dioxide

Q3. The property of a surface that provides a measure of how efficiently a surface emits energy relative to a black body is known as?

(a) Irradiation (c) Absorptivity

(b) Emissive power (d) Emissivity

Q4. The transmissivity of an opaque surface is equal to?

(a) Zero (c) Infinity

(b) One (d) Two

Q5. A surface for which its absorptivity is equal to its emissivity is known as?

(a) Blackbody (c) Grey body

(b) White body (d) Transparent body

Q6.For an opaque, grey surface with emissivity (ε) of 0.38, the reflectivity of the surface is equal to

(a) 0.38 (c) 1.00

(b) 0.62 (d) 0.26

Q7. A 0.5 m^2 opaque grey surface with a temperature of 127 °C has absorptivity of 0.4. What is the surface emissive power in W/m^2?

(a) 581 (b) 675 (c) 871 (d) 775

Q8. The inner and outer surfaces of a 5-m x 6-m brick wall of thickness 30 cm and constant thermal conductivity 0.69 W/m. ^{0}C are maintained at temperatures of 35 ^{0}C and 20 ^{0}C, respectively. The rate of heat transfer through the wall, in W, is

(a) 1035 (b) 1242 (c) 1380 (d) 1898

Q9. Hot air at 80 ^{0}C is blown over a 2-m x 4-m flat surface at 30 ^{0}C. If the average convection heat transfer coefficient is 55 W/m^2 ^{0}C, the rate of heat transfer from the air to the plate, in kW, is (a) 11 (b) 22 (c) 33 (d) 44

Q10. The resistance wire of a 1300-W hair dryer is 80 cm long and has diameter of 0.3 cm. Determine the rate of heat generation in the wire per unit volume, in W/cm^3

(a) 200.3 (b) 212.2 (c) 229.9 (d) 530.9

Q11. A heat flux meter attached to the inner surface of a 3-cm-thick refrigerator door indicates a heat flux of 25 W/m^2 through the door. Also, the temperatures of the inner and outer surfaces of the door are measured to be 7 ^{0}C and 15 ^{0}C, respectively. The average thermal conductivity of the refrigerator door, in W/m ^{0}C, is

(a) 0.0638 (b) 0.0568 (c) 0.0734 (d) 0.0938

Q12. The rate at which radiant energy is incident on a surface per unit area of the surface is known as?

(a) Incident radiation (c) Irradiation

(b) Insolation (d) Emissive power

Q13. All the following are convection heat transfer process except

(a) Boiling
(b) Natural convection
(c) Condensation
(d) None of the above

Q14. Which of the following statement(s) is not true about radiation?

(a) Radiation is usually significant relative to natural convection but negligible relative to forced convection.
(b) Radiation heat transfer to or from a surface surrounded by a gas occurs parallel to convection if there is bulk gas motion between the surface and the gas.
(c) Radiation is usually significant relative to forced convection but negligible relative to natural convection.
(d) All bodies at a temperature above absolute zero emit thermal radiation.

Q15. Which of the following statements about convection is not correct?

(a) Forced convection is always accompanied by natural convection
(b) Heat transfer coefficients encountered in forced convection are typically much higher than those encountered in natural convection
(c) The presence of temperature gradient in a fluid in a gravity field always gives rise to natural convection currents
(d) Free convection heat transfer coefficient for liquids range between 2 and 25 $W/m^2 K$

Q16. The convection heat transfer mode is due to the combined effects of

(a) Convection and bulk fluid motion
(b) Convection and film fluid motion
(c) Conduction and film fluid motion
(d) Conduction and bulk fluid motion

Q17. Which of the following statement(s) is/are correct?

I. The thermal conductivity of constantan (55% Cu, 45% Ni) an alloy of two metals is much lower than the thermal conductivity of either metal of which it is composed.
II. Commercial bronze is an alloy of two metals of thermal conductivities k_1 and k_2. The thermal conductivity k of the alloy is between k_1 and k_2.
III. If a material has uniform properties in all directions, in heat transfer analysis it said to *isotropic*.

(a) II only
(b) II and II only
(c) I and III only
(d) I and II only

Q18. The total radiation leaving a surface per unit time per unit surface area is called:

(a) Emissive power
(b) Emissivity
(c) Irradiation
(d) Radiosity

Q19. Heat transfer through a *vacuum* is by:

(a) Radiation only
(b) Convection only
(c) Convection and conduction only
(d) Convection and radiation only

Q20. The driving potential difference for heat flow is ..... in the direction of heat flow.

(a) Positive
(b) Negative
(c) Undefined
(d) Absolute

UNIT 2

Q1. For radial conduction in a hollow cylindrical wall of inner radius, r_1 and outer radius, r_2, the thermal resistance is of the form;

(a) $\frac{\ln\left(r_2/r_1\right)}{2\pi \cdot kL}$ (b) $\frac{\ln\left(r_1/r_2\right)}{2\pi \cdot kL}$ (c) $\frac{r_2 - r_1}{4\pi \cdot kr_1 r_2}$ (d) $\frac{r_1 - r_2}{4\pi \cdot kr_1 r_2}$

Q2. For radial conduction in a hollow spherical shell of inner radius, r_1 and outer radius, r_2, the thermal resistance is of the form;

(a) $\frac{\ln\left(r_2/r_1\right)}{2\pi \cdot kL}$ (b) $\frac{\ln\left(r_1/r_2\right)}{2\pi \cdot kL}$ (c) $\frac{r_2 - r_1}{4\pi \cdot kr_1 r_2}$ (d) $\frac{r_1 - r_2}{4\pi \cdot kr_1 r_2}$

Q3. In heat transfer analysis, the ratio of the thermal conductivity of material to its heat capacity is termed as

(a) Biot number
(b) Thermal diffusivity
(c) Prandtl number
(d) Thermal capacity

Q4. For a spherical body, the critical radius of insulation is given by

(a) $\frac{h}{2k}$ (b) $\frac{k}{h}$ (c) $\frac{2k}{h}$ (d) $\frac{h}{k}$

Q5. The overall heat transfer coefficient, U of a heat exchanger is 24 W/m^2. °C. For a surface area of 2 m^2, what is the thermal resistance of the heat exchanger?

(a) 0.0833 °C/W
(b) 0.0208 °C/W
(c) 0.0417 °C/W
(d) 0.0112 °C/W

Q6. Water at 80 ^{0}C flows through a 50 mm bore steel pipe of 6 mm thickness, and the atmospheric temperature is 15 ^{0}C. The thermal conductivity of the steel is 48 W/m K and the inside and outside heat transfer coefficients are 2800 and 17 W/m^2 K, respectively. Neglecting radiation, calculate the rate of heat loss per unit length of pipe.

(a) 325 W/m
(b) 213 W/m
(c) 420 W/m
(d) 516 W/m

Q7. A brick wall of thickness 20 cm and thermal conductivity 0.7 W/ (m °C) is maintained at 25°C at one surface and 4°C at the other surface. The heat flow rate (**in Watt**) across a 4-m^2 surface area of the wall is

(a) 138
(b) 294
(c) 2.94
(d) 1.38

Q8. A fibre-glass insulating board with thermal conductivity 0.08 W/ (m °C) is to be used to limit the heat losses to 80 W/m^2 for a temperature difference of 100°C across the board. The thickness (**in mm**) of the insulating board is:

(a) 100
(b) 75
(c) 0.1
(d) 0.075

Use the information below to answer questions 9 to 14

For the composite wall shown in Figure Q9-14, thermal conductivities are: $k_1 = k_3 = 80$ W/ (m. K), $k_2 = 120$ W/ (m. K), $k_4 = 100$ W/ (m. K), and $k_5 = 150$ W/ (m. K). Air temperature is 30 ºC on the left side and 80 ºC on the right side. Convection coefficient is h = 15 W/ (m^2. K) on both sides.

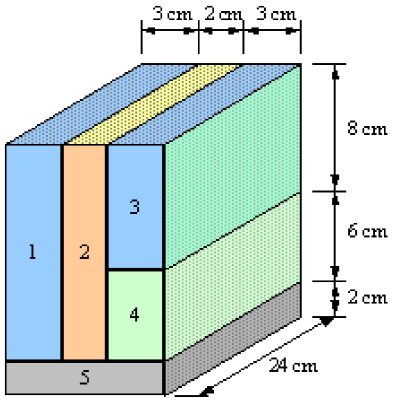


Figure Q 9-14

Q9. The thermal resistance in (**K/W**) for layer 1 is

(a) 0.01008
(b) 0.01116
(c) 0.02083
(d) 0.02120
(e) 0.1111

Q10. The thermal resistance in (**K/W**) for layer 4 is

(a) 0.01008
(b) 0.01116
(c) 0.02083
(d) 0.02120
(e) 0.1111

Q11. The thermal resistance in (**K/W**) for layer 5 is

(a) 0.01008
(b) 0.01116
(c) 0.02083
(d) 0.02120
(e) 0.1111

Q12. The equivalent thermal resistance in (**K/W**) for layers 3 and 4 in combination is

(a) 0.01008
(b) 0.01116
(c) 0.02083
(d) 0.02120
(e) 0.1111

Q13. The total thermal resistance for the composite wall in (**K/W**) is

(a) 0.01008
(b) 0.01116
(c) 0.02083
(d) 0.02120
(e) 0.1111

Q14. The rate of heat transfer in (**W**) through the composite wall is

(a) 14.31
(b) 450.05
(c) 2358.49
(d) 4480.29
(e) 4960.32

Q15. The performance of a fin attached to a surface can be expressed in terms of its effectiveness. If the effectiveness of the fin is ($\varepsilon = 1$), it indicates that the

(a) Fin is enhancing heat transfer from the primary surface
(b) Fin is actually acting as an insulator
(c) Addition of the fin to the surface does not affect heat transfer at all
(d) Thermal conductivity of the fin is very low

Conduction with internal Heat generation Application of Fourier's law: Use the information below to answer questions 16 - 18.

The temperature distribution across a wall 1 m thick at a certain instant of time is given as

$$T(x) = 900 - 300x - 50x^2,$$

where, T is in degree Celsius and x is in metres. A uniform internal heat generation, $\dot{q} = 1000$ W/m^3, is present in the wall of area 10 m^2 having constant properties ρ = 1600 kg/m^3, k = 40 W/ (m K), and $c_p = 4$ (kJ/kg.K). Assume that there is one-dimensional heat conduction in the positive x-direction.

Q16. The rate of heat transfer entering the wall ($x = 0$) is

(a) 160 kW
(b) 120 kW
(c) 100 kW
(d) 90 kW
(e) 360 kW

Q17. The rate of heat transfer leaving the wall ($x = 1$m) is

(a) 160 kW
(b) 120 kW
(c) 100 kW
(d) 90 kW
(e) 220 kW

Q18. The rate of energy storage in the wall is

(a) – 30 kW
(b) – 40 kW
(c) – 10 kW
(d) – 15 kW
(e) – 50 KW

Q19. Which of the following is the correct expression for the efficiency of a fin of length L with negligible heat loss from the fin tip?

(a) $\eta_{fin} = mL$
(b) $\eta_{fin} = \tanh mL$
(c) $\eta_{fin} = \dfrac{\tanh mL}{mL}$
(d) $\eta_{fin} = \dfrac{1}{mL}$
(e) $\eta_{fin} = \dfrac{mL}{\tanh mL}$

Q20. The performance of a fin attached to a surface can be expressed in terms of its effectiveness, ε_{fin}. If the effectiveness of the fin is greater than unity, That is, $\varepsilon_{fin} > 1$, indicates that the

(a) Fin is enhancing heat transfer from the primary surface
(b) Fin is actually acting as an insulator
(c) Addition of the fin to the surface does not affect heat transfer at all
(d) Thermal conductivity of the fin is very low
(e) Fin is long

UNIT 3

Q1. A small flat plate coated with white paint is suspended inside a large enclosure, and its temperature is maintained at 400 K. The surface of the enclosure is maintained at 3000 K and the spectral distribution of its emissivity has the following characteristics:

ε_λ	0.2	0.7
λ (μm)	< 2.0	> 2.0

The total emissivity, ε of the enclosure surface is approximately

(a) 0.428
(b) 0.331
(c) 0.523
(d) 0.459
(e) 0.518

Q2. The wavelength at which the blackbody spectral emissive power $E_{b\lambda}$ is a maximum for a blackbody at 2000 K falls within which portion of the electromagnetic spectrum?

(a) Visible light (c) Infrared (e) Microwaves
(b) Ultraviolet (d) X-rays

Q3. The temperature of the blackbody such that 65 percent of the energy emitted by it lie in the wavelength spectrum $\lambda = 0$ and 10 μm is

(a) 411.1 K (c) 641.1 K (e) 4200 K
(b) 513.9 K (d) 5128.9 K

Q4. The reflectivity of a surface averaged over all directions and at a specified wavelength is known as

(a) Total directional reflectivity (d) Hemispherical reflectivity
(b) Directional reflectivity (e) Total hemispherical reflectivity
(c) Spectral hemispherical reflectivity

Q5. Materials can exhibit different behaviour at different wavelengths. Which of the following is true for materials such as glass and water?

(a) They are opaque to visible radiation and transparent to infrared radiation
(b) They are transparent to visible radiation and opaque to ultraviolet radiation
(c) They are transparent to visible radiation and opaque to infrared radiation
(d) They are opaque to ultraviolet radiation and transparent to infrared radiation
(e) They are transparent to ultraviolet radiation and opaque to infrared radiation

Q6. The incident solar radiant flux at the earth's mean orbital radius of 1.4967×10^{11} m from the sun is $G = 1402.6$ W/m^2. Taking the diameter of the sun, D to be 1.39×10^9 m, the effective surface temperature of the sun is approximately

(a) 6000 K (c) 5762 K (e) 4875 K
(b) 5820 K (d) 5020 K

Q7. An enclosure consists of two surfaces. The view factor from surface 1 to surface 2 is 0.25. If the area of surface 2 is two-thirds that of surface 1, then the view factor from surface 2 to surface 1 is

(a) 0.431 (c) 0.158 (e) 0.235
(b) 0.375 (d) 0.167

Q8. Consider an enclosure consisting of 8 surfaces. The total number of view factors of the geometry that can be determined by the application of the reciprocity and summation rule is

(a) 36 (c) 8 (e) 64
(b) 28 (d) 16

Q9. Consider heat exchange between two horizontal infinitely long parallel plates. The emissivities of the top and the bottom plates are 0.5 and 0.8, respectively. The top (surface 2) and the bottom (surface 1) plates are maintained at 300 and 400 K, respectively. The net radiation heat flux from the bottom plate to the top plate, in W/ m^2, is

(a) 325.2 (c) 233.5 (e) 467.0
(b) 105.9 (d) 441.0

Q10. An empty 'ideal' Milo tin insulated on the outside surfaces can be considered as a three surface enclosure. The inner surface of the top cover is designated as surface 1; the inner surface of bottom cover is designated as surface 2; and the inner surface of the curved part is designated as surface 3. Each of the top and bottom covers is of area 4 m^2 and the inner curved surface is of area 6 m^2. If the view factor from surface 1 to surface 2 is 0.28 (i.e.$F_{12} = 0.28$), F_{31} is

(a) 0.60 (c) 0.72 (e) 0.86
(b) 0.48 (d) 0.28

Q11. Radiation heat transfer between two surfaces can be greatly reduced by inserting thin plates or shields between the two surfaces. Such thin plates or shields should be materials of

(a) High-emissivity (d) Low-absorptivity (e) High-absorptivity
(b) Low-reflectivity
(c) Low-emissivity

Q12. Consider heat exchange between two horizontal infinitely long parallel plates. The emissivities of both the top and the bottom plates are 0.5. The top (surface 2) and the bottom (surface 1) plates are maintained at 300 and 400 K respectively. Determine the net radiation heat flux between the plates, in W/ m^2, when a thin aluminium sheet with an emissivity of 0.5 is placed between the plates.

(a) 330.75 (c) 233.5 (e) 165.38
(b) 105.9 (d) 300.3

Q13. An enclosure consists of N surfaces. The total number of view factors of the enclosure that can be determined directly is given by which of the following expressions?

(a) N^2
(b) N
(c) $\frac{1}{2}N(N-1)$
(d) $\frac{1}{2}N^2(N-1)$
(e) $\frac{1}{2}N(N+1)$

Q14. Microwaves are very suitable for use in cooking since they are reflected by metals, transmitted by glasses and plastics, absorbed by food. Which of the gives the range of wavelength of microwaves?

(a) 0.40 – 0.76 μm (c) $10^2 - 10^5$ μm (e) 1 – 5 μm
(b) $10^5 - 10^{10}$ μm (d) $0.1 - 10^2$ μm

Q15. An enclosure consists of N surfaces. The total number of view factors of the enclosure that can be determined by the summation rule is given by which of the following expressions?

(a) N^2
(b) N
(c) $\frac{1}{2}N(N-1)$
(d) $\frac{1}{2}N^2(N-1)$
(e) $\frac{1}{2}N(N+1)$

Q16. The view factor from the base of a cube to each of the sides given that the view factor from the base to the top is 0.2 is:

(a) 1/6 (b) 1/5 (c) 1/4 (d) 1/3 (e) 1/2

Q17. According to Wein's displacement law, the maximum emissive power of a black body at a temperature of 727 ^{0}C occurs at a wavelength of approximately

(a) 2.9 μm (c) 4.5 μm (e) 7.4 μm
(b) 3.7 μm (d) 5.8 μm

Q18. In radiation heat transfer, a blackbody is said to be a *diffuse* emitter. The term *diffuse* means

(a) Dependent on wavelength
(b) Independent of wavelength
(c) Independent of direction
(d) Dependent on direction

Q19. In radiation heat transfer, the term *spectral* indicates dependence on:

(a) Temperature
(b) Direction
(c) Wavelength
(d) Pressure
(e) Surface finish

Q20. The variation of black emissive power with wavelength for several temperatures is shown in Figure Q20. Which of the following conclusions is true of the diagram?

(a) At any wavelength, the amount of emitted radiation decreases with increasing temperature
(b) A larger fraction of the radiation is emitted at shorter wavelengths at higher temperatures
(c) The emitted radiation is a discrete function of wavelength.
(d) As temperature increases, the peak of the curve shifts towards longer wavelengths
(e) A larger fraction of the radiation is emitted at shorter wavelengths at lower temperatures

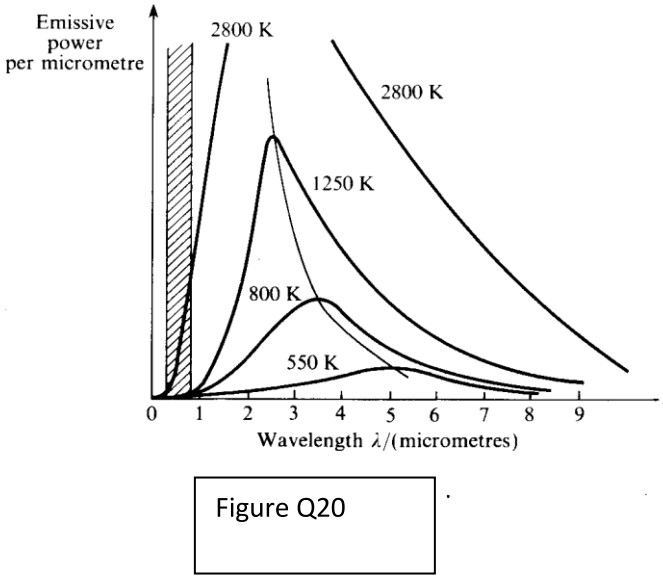


Figure Q20

Q21. The total radiation leaving a surface per unit time per unit surface area is called:

(a) Emissive power
(b) Emissivity
(c) Hemispherical emission
(d) Irradiation
(e) Radiosity

UNIT 4

Q1. For flow over a flat plate by forced convection mode, transition from laminar flow to turbulent flow occurs at which critical Reynolds number?

(a) 2300 (c) 5×10^5 (e) 2000
(b) 1×10^9 (d) 2×10^3

Q2. The Pecklet number, Pe, is a dimensionless number relevant in the study of transport phenomena in fluid flow. It is defined as

(a) Gr/Re^2
(b) $Ra \cdot Pr$
(c) $Re \cdot Pr$
(d) $Gr \cdot Pr$
(e) Pr/Re^2

Q3. The Grashof number is the ratio of

(a) Inertia forces to viscous forces
(b) Elastic forces to viscous forces
(c) Buoyancy forces to inertia forces
(d) Buoyancy forces to viscous forces
(e) Inertia forces to buoyancy forces

Q4. For flow in a circular tube by forced convection mode, transition from laminar flow to turbulent flow occurs at which critical Reynolds number?

(a) 2.3×10^3 (c) 5.0×10^5 (e) 2.1×10^4
(b) 1.0×10^9 (d) 5.0×10^3

Q5. The Rayleigh number, Ra, is a dimensionless number associated with buoyancy driven flow. It is defined as

(a) Gr/Re^2
(b) $Nu \cdot Pr$
(c) $Re \cdot Pr$
(d) $Gr \cdot Pr$
(e) Pr/Re^2

Q6. The Reynolds number is the ratio of

(a) Inertia forces to viscous forces
(b) Elastic forces to viscous forces
(c) Buoyancy forces to inertia forces
(d) Buoyancy forces to viscous forces
(e) Inertia forces to buoyancy forces

Use the following information to answer questions 7-10

Air at 14 °C and 1 atm pressure flows over a 5-m-long flat plate whose temperature is 40 °C with a velocity of 3 m/s. for forced convection over a flat plate, take the average Nusselt number as;

$Nu = 0.664\, Re_L^{0.5} \cdot Pr^{1/3}$ For laminar flow

$Nu = 0.037\, Re_L^{0.8} \cdot Pr^{1/3}$ For turbulent flow

The properties of air at the film temperature are;

$k = 0.026 W/m.^{\circ}C, \rho = 1.177 kg/m^3, Pr = 0.712, \upsilon = 1.57 \times 10^{-5}\ m^2/s$

Q7. Determine the Reynolds number at the end of the plate.

(a) 4.13×10^4 (c) 1.10×10^4 (e) 9.60×10^5
(b) 2.38×10^5 (d) 3.43×10^4

Q8. Determine the type of flow and the corresponding average Nusselt number

(a) Laminar flow; 2018
(b) Turbulent flow; 4130
(c) Laminar flow; 1918
(d) Turbulent flow; 2018
(e) Turbulent flow; 1918

Q9. Determine the rate of heat transfer per unit width of the entire plate

(a) 4,936 W (c) 2,792 W (e) 8,330 W
(b) 1,364 W (d) 1,297 W

Q10. Determine the specific heat capacity of the air

(a) 1.964 kJ/ kg. °C (d) 2.047 kJ/ kg. °C
(b) 1.880 kJ/ kg. °C (e) 2.219 kJ/ kg. °C
(c) 1.002 kJ/ kg. °C

Q11. For forced convection heat transfer analysis, the Nusselt number is a function of

(a) Rayleigh number and Prandtl number
(b) Reynolds number and Prandtl number
(c) Grashof number and Reynolds number
(d) Grashof number and Prandtl number
(e) Rayleigh number and Reynolds number

Q12. The presence of a temperature gradient in a fluid in a gravity field always gives rise to

(a) Forced convection currents (d) Ocean currents
(b) Natural convection currents (e) Opposing current
(c) Acceleration currents

Q13. Which of the following parameters represent the importance of natural convection relative to forced convection?

(a) Gr/Re^2 (c) Gr/Ra^2 (e) Re/Pr^2
(b) Re/Gr^2 (d) Ra/Gr^2

Q14. The convection heat transfer coefficient is a strong function of which of these numbers in forced convection?

(a) Prandtl number (d) Grashof number
(b) Rayleigh number (e) Nusselt number
(c) Reynolds number

Q15. The convection heat transfer coefficient is a strong function of which of these numbers in natural convection?

(a) Prandtl number (d) Grashof number
(b) Rayleigh number (e) Nusselt number
(c) Reynolds number

Q16. For heat transfer in a given fluid, the parameter Gr/Re^2 represents the importance of natural convection to forced convection. Which of the following statement(s) is/are correct?

I. Natural convection is negligible when $Gr/Re^2 < 0.1$
II Forced convection is negligible when $Gr/Re^2 > 10$
III. Neither natural convection nor forced convection is negligible when
$0.1 < Gr/Re^2 < 10$.

(a) I only (d) I and III only
(b) II only (e) I, II and III
(c) I and II only

Q17. For flow over a vertical plate by natural convection mode, transition from laminar flow to turbulent flow occurs at which critical Grashof number?

(a) 2300 (c) 5×10^5 (e) 2000
(b) 1×10^9 (d) 2×10^3

Q18. Which one of the following dimensionless parameters is itself a property?

(a) Reynolds number
(b) Prandtl number
(c) Grashof number
(d) Rayleigh number
(e) Nusselt number

Q19. For Natural convection heat transfer analysis, the Nusselt number is a function of

(a) Rayleigh number and Prandtl number
(b) Reynolds number and Prandtl number
(c) Grashof number and Reynolds number
(d) Grashof number and Prandtl number
(e) Rayleigh number and Reynolds number

Q20. The convection heat transfer coefficient can be determined from which of the following dimensionless parameters?

(a) Reynolds number
(b) Prandtl number
(c) Grashof number
(d) Rayleigh number
(e) Nusselt number

UNIT 5

Heat exchanger analysis using the effectiveness - NTU method

Use the information to answer questions 1-3

Cold water at 20 °C (c_p = 4200 J/kg. °C) enters the thinned walled tubes of a heat exchanger with 2-shell passes and 20-tube passes at a rate of 3 kg/s with a tube diameter of 1.5 cm and leaves at 65 °C, while Hot oil (c_p = 2100 J/kg. °C) enters the shell side at 125 °C at the same mass flow rate. The overall heat transfer coefficient based on the outer surface of the tube is 1250 W/ m^2 °C and the heat transfer surface area on that side is 6.5 m^2.

Q1. The length of each tube, (in cm) of this multi-pass heat exchanger is

(a) 690.0
(b) 137.9
(c) 345.0
(d) 17.31
(e) 22.4

Q2. The maximum possible rate of heat transfer in kW in this multi-pass heat exchanger is

(a) 661.5
(b) 378.0
(c) 1323.0
(d) 756.1
(e) 882.0

Q3. The heat transfer effectiveness of this multi-pass heat exchanger are;

A. 80.7 percent
B. 57.1 percent
C. 85.7 percent
D. 60.5 percent
E. 85.7 percent

Q4. For specified inlet and outlet temperatures, the logarithmic mean temperature difference for a counter flow heat exchanger is

(a) Equal to that of a parallel flow heat exchanger
(b) Less than that of a cross flow heat exchanger
(c) Always greater than that of a parallel flow heat exchanger
(d) Twice that of a parallel flow heat exchanger
(e) Less than that of a multi-pass heat exchanger

Q5. An adiabatic heat exchanger is used to heat cold water at 20 °C entering at a rate of 4 kg/s by hot water at 90 °C entering at a rate of 3 kg/s. If the exit temperature of the hot water is 50 °C and the c_p of water is 4.18 kJ/kg °C, the exit temperature of the cold water is

(a) 60 °C (b) 73 °C (c) 50 °C (d) 55 °C (e) 40 °C

Q6. The fouling factor depends on the operating temperature and the velocity of the fluids as well as the length of service. It increases with

(a) Increasing temperature and increasing velocity
(b) Decreasing temperature and increasing velocity
(c) Increasing temperature and decreasing velocity
(d) Decreasing temperature and decreasing velocity
(e) Increasing temperature and constant velocity

Q7. The flow configuration in compact heat exchangers is usually referred to as?

(a) Parallel flow (c) Couette flow (e) Longitudinal flow
(b) Counter flow (d) Cross flow

Q8. For specified inlet and outlet temperatures, the correction factor, F, for a multi-pass heat exchanger is

(a) Equal to that of a parallel flow heat exchanger
(b) Less than that of a cross flow heat exchanger
(c) Always greater than that of a parallel flow heat exchanger
(d) Twice that of a parallel flow heat exchanger
(e) Less than that of a counter heat exchanger

Heat exchanger analysis using the logarithmic mean temperature difference (LMTD) method

Use the information to answer questions 9-11

Steam in the condenser of a steam power plant is to be condensed at a temperature of 40 °C with cooling water from a nearby lake, which enters the tubes of the condenser at 16 °C and leaves at 28 °C. The surface area of the tubes is 35 m^2, and the overall heat transfer coefficient is 2100 W/m^2. °C. Take the specific enthalpy of vaporization (h_{fg}) of water at 40 °C to be 2395 kJ/kg. K and the specific heat capacity of cold water to be 4182 J/ kg. K.

Q9. The logarithmic mean temperature difference in °C, along the length of the exchanger is

(a) 10.62 (c) 20.86 (e) 22.4
(b) 11.50 (d) 17.31

Q10. The rate of heat transfer in the condenser, in kW, is determined to be

(a) 1272.5 (c) 1872.5 (e) 882.0
(b) 1087.1 (d) 579.6

Q11. The mass flow rate of the cooling water needed and the rate of condensation of the steam in the condenser are;

(a) 32.50 kg/s and 0.45 kg/s, respectively
(b) 25.36 kg/s and 0.53 kg/s, respectively
(c) 17.25 kg/s and 0.81 kg/s, respectively
(d) 12.50 kg/s and 0.45 kg/s, respectively
(e) 32.5 kg/s and 0.53 kg/s, respectively

Q12. For a given NTU and capacity ratio, which of the following heat exchangers has the highest effectiveness?

(a) One shell pass and n- tube passes
(b) Cross flow with one fluid mixed and the other fluid unmixed
(c) Counter flow heat exchanger
(d) Parallel flow heat exchanger
(e) Cross flow heat exchanger with both fluid unmixed

Q13. The heat capacity rate of a fluid during a phase change process is:

(a) Infinite (c) One (e) Partially
(b) Zero (d) Indeterminate constant

Q14. Regardless of the configuration of the heat exchanger, the correction factor for a condenser is:

(a) Indeterminate (c) One (e) less than one
(b) Zero (d) More than one

Q15. Which of the following heat exchangers enables a direct contact between a liquid and a gas?

(a) Regenerators (d) Multi- pass heat exchanger
(b) Recuperator (e) Shell- and- tube heat exchanger
(c) Evaporative heat exchanger

Q16. The correction factor, F, depends on the inlet and outlet temperatures of the hot and cold fluids as well as ……..

(a) The type of fluid in the heat exchanger
(b) The geometry of the heat exchanger
(c) The surface area of the heat exchanger
(d) The capacity of the heat exchanger
(e) The length of the heat exchanger

Q17. A heat exchanger is to cool oil (C_p = 2200 J/kg.^{o}C) at a rate of 20 kg/s from 120 ^{o}C to 50 ^{o}C by air. The heat transfer rating of the heat exchanger is:

(a) 110 kW (c) 2,200 kW (e) 44 kW
(b) 3,080 kW (d) 6,000 kW

Q18. Hot oil is to be cooled by water in a shell-and-tube heat exchanger. The heat capacity rates of the hot and cold fluids are 6 and 8 kW/K, respectively. The overall heat transfer coefficient and the heat transfer surface area are 312.5 W/ (m^2 K) and 12 m^2, respectively. The number of transfer units (NTU) for this exchanger is

(a) 1.600 (c) 0.200 (e) 0.025
(b) 0.625 (d) 0.125

Heat exchanger analysis by effectiveness-NTU method.
Use the information below to answer questions 19 – 21.

An engine oil cooler consists of a single-pass, counter-flow shell-and-tube heat exchanger with 300 tubes of internal diameter 7.3 mm and length 8 m. The oil [C_p = 3420 J/ (kg .^{0}C)] flows in the tube side entering at a mass flow rate of 8 kg/s at a temperature of 70 ^{0}C. Cooling water [C_p = 4200 J/ (kg .^{0}C)] in the shell side enters at a mass flow rate of 12 kg/s at a temperature of 15 ^{0}C. The overall heat transfer coefficient is 546.6 W/ (m^2. ^{0}C) and the effectiveness of the exchanger is 58.8 %.

Q19. The number of transfer units for this heat exchanger is

(a) 8.0 (b) 4.4 (c) 2.0 (d) 1.2 (e) 1.1

Q20. The thermal capacity ratio for this heat exchanger is

(a) 0.543 (c) 0.667 (e) 0.819
(b) 0.588 (d) 0.814

Q21. The outlet temperature of the oil is

(a) 10.4 ^{0}C (c) 32.3 ^{0}C (e) 52.4 ^{0}C
(b) 27.5 ^{0}C (d) 37.7 ^{0}C

Q22. Steam in the condenser of a steam power plant is to be condensed at a temperature of 40 ^{0}C with cooling water from a nearby lake, which enters the tubes of the condenser at 14 ^{0}C and leaves at 22 ^{0}C. The logarithmic mean temperature difference, in ^{0}C, along the length of the exchanger is approximately

(a) 11 (c) 13 (e) 22
(b) 12 (d) 19

Q23. A heat exchanger is used to heat cold water (C_p = 4.18 kJ/kg K) at 15 ^{0}C entering at a rate of 2 kg/s, by hot air (Cp = 1.01 kJ/kg/K) at 100 ^{0}C entering at a rate of 3 kg/s. If the exit temperature of hot air is 20 ^{0}C, the exit temperature of cold water is:

(a) 44 ^{0}C
(b) 49 ^{0}C
(c) 39 ^{0}C
(d) 72 ^{0}C
(e) 95 ^{0}C

SUGGESTED ANSWERS

UNIT	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
# 1	**B**	**D**	**D**	**A**	**C**	**B**	**A**	**A**	**B**	**C**	**D**	**C**	**D**	**C**	**D**	**D**	**C**	**D**	**A**	**B**	**-**	**-**	**-**
# 2	**A**	**C**	**B**	**C**	**B**	**B**	**B**	**A**	**B**	**C**	**E**	**A**	**D**	**A**	**C**	**B**	**A**	**A**	**C**	**A**	**-**	**-**	**-**
# 3	**B**	**C**	**B**	**C**	**C**	**B**	**B**	**A**	**D**	**B**	**C**	**E**	**E**	**C**	**B**	**B**	**A**	**C**	**C**	**B**	**E**	**-**	**-**
# 4	**C**	**C**	**D**	**A**	**D**	**A**	**E**	**D**	**C**	**C**	**B**	**B**	**A**	**C**	**D**	**E**	**B**	**B**	**D**	**E**	**-**	**-**	**-**
# 5	**C**	**A**	**C**	**C**	**C**	**C**	**D**	**E**	**D**	**B**	**B**	**C**	**A**	**C**	**C**	**B**	**B**	**B**	**E**	**B**	**D**	**A**	**A**